

EXERCISES 3.3

Exer. 1–40: Find the derivative.

1 $g(t) = 6t^{5/3}$

2 $h(z) = 8z^{3/2}$

3 $f(s) = 15 - s + 4s^2 - 5s^4$

4 $f(t) = 12 - 3t^4 + 4t^6$

5 $f(x) = 3x^2 + \sqrt[3]{x^4}$

6 $g(x) = x^4 - \sqrt[4]{x^3}$

7 $g(x) = (x^3 - 7)(2x^2 + 3)$

8 $k(x) = (2x^2 - 4x + 1)(6x - 5)$

9 $f(x) = x^{1/2}(x^2 + x - 4)$

10 $h(x) = x^{2/3}(3x^2 - 2x + 5)$

11 $h(r) = r^2(3r^4 - 7r + 2)$

12 $k(v) = v^3(-2v^3 + v - 3)$

13 $g(x) = (8x^2 - 5x)(13x^2 + 4)$

14 $H(z) = (z^5 - 2z^3)(7z^2 + z - 8)$

15 $f(x) = \frac{4x - 5}{3x + 2}$

16 $h(x) = \frac{8x^2 - 6x + 11}{x - 1}$

17 $h(z) = \frac{8 - z + 3z^2}{2 - 9z}$

18 $f(w) = \frac{2w}{w^3 - 7}$

19 $G(v) = \frac{v^3 - 1}{v^3 + 1}$

20 $f(t) = \frac{8t + 15}{t^2 - 2t + 3}$

21 $g(t) = \frac{\sqrt[3]{t^2}}{3t - 5}$

22 $f(x) = \frac{\sqrt{x}}{2x^2 - 4x + 8}$

23 $f(x) = \frac{1}{1 + x + x^2 + x^3}$

24 $p(x) = 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$

25 $h(x) = \frac{7}{x^2 + 5}$

26 $k(z) = \frac{6}{z^2 + z - 1}$

27 $F(t) = t^2 + \frac{1}{t^2}$

28 $s(x) = 2x + \frac{1}{2x}$

29 $K(s) = (3s)^{-4}$

30 $W(s) = (3s)^4$

31 $h(x) = (5x - 4)^2$

32 $S(w) = (2w + 1)^3$

33 $g(r) = (5r - 4)^{-2}$

34 $S(x) = (3x + 1)^{-2}$

35 $f(t) = \frac{3/(5t) - 1}{(2/t^2) + 7}$

36 $N(z) = \frac{4/z^2}{(3/z) + 2}$

37 $M(x) = \frac{2x^3 - 7x^2 + 4x + 3}{x^2}$

38 $T(z) = \frac{5z^4 + z^3 - 2z}{z^3}$

39 $f(x) = \frac{3x^2 - 5x + 8}{7}$

40 $h(t) = \frac{3t^5 + 2t}{5}$

Exer. 41–44: Solve the equation $D_x y = 0$.

41 $y = 2x^3 - 3x^2 - 36x + 4$

42 $y = 4x^3 + 21x^2 - 24x + 11$

43 $y = \frac{2x^2 + 3x - 6}{x - 2}$

44 $y = \frac{x^2 + 2x + 5}{x + 1}$

Exer. 45–46: Solve the equation $D_x^2 y = 0$.

45 $y = 6x^4 + 24x^3 - 540x^2 + 7$

46 $y = 6x^5 - 5x^4 - 30x^3 + 11x$

Exer. 47–50: Find dy/dx by (a) using the quotient rule, (b) using the product rule, and (c) simplifying algebraically and using (3.18).

47 $y = \frac{3x - 1}{x^2}$

48 $y = \frac{x^2 + 1}{x^4}$

49 $y = \frac{x^2 - 3x}{\sqrt[3]{x^2}}$

50 $y = \frac{2x + 3}{\sqrt{x^3}}$

Exer. 51–52: Find d^2y/dx^2 .

51 $y = \frac{3x + 4}{x + 1}$

52 $y = \frac{x + 3}{2x + 3}$

Exer. 53–54: Find an equation of the tangent line to the graph of f at P .

53 $f(x) = \frac{5}{1 + x^2}$; $P(-2, 1)$

54 $f(x) = 3x^2 - 2\sqrt{x}$; $P(4, 44)$

55 Find the x -coordinates of all points on the graph of $y = x^3 + 2x^2 - 4x + 5$ at which the tangent line is (a) horizontal (b) parallel to the line $2y + 8x = 5$ 56 Find the point P on the graph of $y = x^3$ such that the tangent line at P has x -intercept 4.57 Find the points on the graph of $y = x^{3/2} - x^{1/2}$ at which the tangent line is parallel to the line $y - x = 3$.58 Find the points on the graph of $y = x^{3/3} + x^{1/3}$ at which the tangent line is perpendicular to the line $2y + x = 7$.

Exer. 59–60: Sketch the graph of the equation and find the vertical tangent lines.

59 $y = \sqrt{x} - 4$

60 $y = x^{1/3} + 2$

61 A weather balloon is released and rises vertically such that its distance $s(t)$ above the ground during the first 10 seconds of flight is given by $s(t) = 6 + 2t + t^2$, where $s(t)$ is in feet and t is in seconds. Find the velocity of the

balloon at

(a) $t = 1$, $t = 4$, and $t = 8$

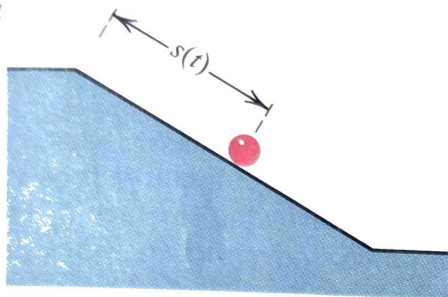
(b) the instant the balloon is 50 feet above the ground

- 62 A ball rolls down an inclined plane such that the distance (in centimeters) that it rolls in t seconds is given by $s(t) = 2t^3 + 3t^2 + 4$ for $0 \leq t \leq 3$ (see figure).

(a) Find the velocity of the ball at $t = 2$.

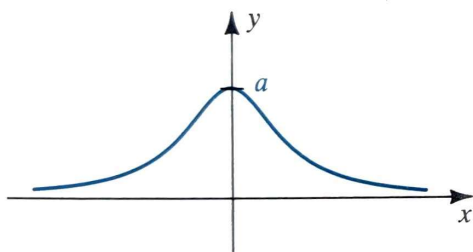
(b) At what time is the velocity 30 cm/sec?

EXERCISE 62

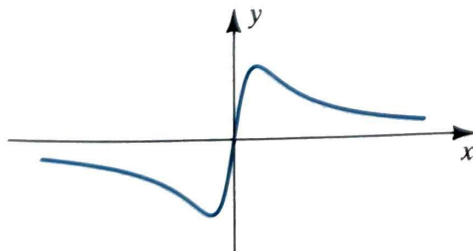


Exer. 63–64: An equation of a classical curve and its graph are given for positive constants a and b . (Consult books on analytic geometry for further information.) Find the slope of the tangent line at the point P .

- 63 Witch of Agnesi: $y = \frac{a^3}{a^2 + x^2}$; $P(a, a/2)$



- 64 Serpentine curve: $y = \frac{abx}{a^2 + x^2}$; $P(a, b/2)$



Exer. 65–66: Find equations of the lines through P that are tangent to the graph of the equation.

- 65 $P(5, 9)$; $y = x^2$

- 66 $P(3, 1)$; $xy = 4$

Exer. 67–70: If f and g are functions such that $f(2) = 3$, $f'(2) = -1$, $g(2) = -5$, and $g'(2) = 2$, evaluate the expression.

- 67 (a) $(f + g)'(2)$ (b) $(f - g)'(2)$ (c) $(4f)'(2)$
 (d) $(fg)'(2)$ (e) $(f/g)'(2)$ (f) $(1/f)'(2)$

- 68 (a) $(g - f)'(2)$ (b) $(g/f)'(2)$

- (c) $(4g)'(2)$ (d) $(ff)'(2)$

- 69 (a) $(2f - g)'(2)$ (b) $(5f + 3g)'(2)$

- (c) $(gg)'(2)$ (d) $\left(\frac{1}{f+g}\right)'(2)$

- 70 (a) $(3f - 2g)'(2)$ (b) $(5g)'(2)$

- (c) $(6f)'(2)$ (d) $\left(\frac{f}{f+g}\right)'(2)$

- 71 If f , g , and h are differentiable, use the product rule to prove that

$$D_x [f(x)g(x)h(x)] = f(x)g(x)h'(x) + f(x)g'(x)h(x) + f'(x)g(x)h(x).$$

As a corollary, let $f = g = h$ to prove that

$$D_x [f(x)]^3 = 3[f(x)]^2 f'(x).$$

- 72 Extend Exercise 71 to the derivative of a product of four functions, and then find a formula for $D_x [f(x)]^4$

Exer. 73–76: Use Exercise 71 to find dy/dx .

- 73 $y = (8x - 1)(x^2 + 4x + 7)(x^3 - 5)$

- 74 $y = (3x^4 - 10x^2 + 8)(2x^2 - 10)(6x + 7)$

- 75 $y = x(2x^3 - 5x - 1)(6x^2 + 7)$

- 76 $y = 4x(x - 1)(2x - 3)$

- 77 As a spherical balloon is being inflated, its radius r (in centimeters) after t minutes is given by $r = 3\sqrt[3]{t}$ for $0 \leq t \leq 10$. Find the rate of change for each of the following with respect to t at $t = 8$:

(a) the radius r

(b) the volume V of the balloon

(c) the surface area S of the balloon

- 78 The volume V (in ft^3) of water in a small reservoir during spring runoff is given by $V = 5000(t + 1)^2$ for t in months and $0 \leq t \leq 3$. The rate of change of volume with respect to time is the instantaneous flow rate into the reservoir. Find the flow rate at times $t = 0$ and $t = 2$. What is the flow rate when the volume is $11,250 \text{ ft}^3$?

- 79 A stone is dropped into a pond, causing water waves that form concentric circles. If, after t seconds, the radius of one of the waves is $40t$ centimeters, find the rate of change, with respect to t , of the area of the circle caused by the wave at

- (a) $t = 1$ (b) $t = 2$ (c) $t = 3$

- 80 Boyle's law for confined gases states that if the temperature remains constant, then $pV = c$, where p is the pressure, V is the volume, and c is a constant. Suppose that at time t (in minutes) the pressure is $20 + 2t$ centimeters of mercury for $0 \leq t \leq 10$. If the volume is 60 cm^3 at $t = 0$, find the rate at which the volume is changing with respect to t at $t = 5$.

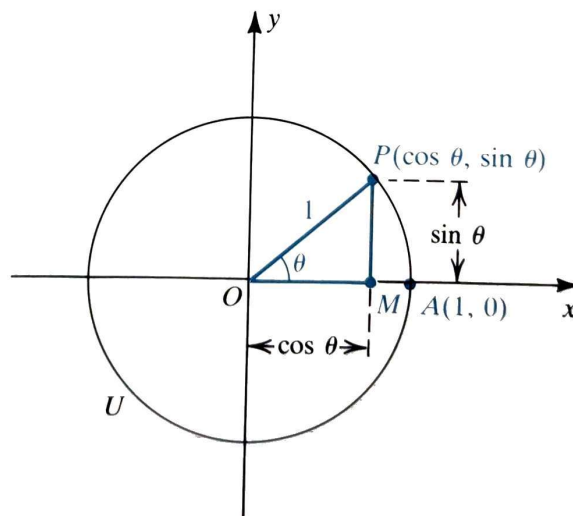
- c 81 (a)** If $f(x) = x^3 - 2x + 2$, approximate $f'(1)$ using Exercise 51 of Section 3.2 with $h = 0.1$.
- (b)** Graph the following on the same coordinate axes: $y = f(x)$, the secant line l_1 through $(1, f(1))$ and $(1.1, f(1.1))$, and the secant line l_2 through $(0.9, f(0.9))$ and $(1.1, f(1.1))$.
- (c)** Find $f'(1)$ and explain why the slope of l_2 is a better approximation to $f'(1)$ than is the slope of l_1 .
- c 82 (a)** If $f(x) = x^{2/3} + 1$, approximate $f'(0)$ using Exercise 51 of Section 3.2 with $h = 0.1$.
- (b)** Graph the following on the same coordinate axes: $y = f(x)$, the secant line through $(0, f(0))$ and $(0.1, f(0.1))$, and the secant line through $(-0.1, f(-0.1))$ and $(0.1, f(0.1))$.
- (c)** Why don't the slopes of the secant lines in (b) approximate $f'(0)$?

3.4 DERIVATIVES OF THE TRIGONOMETRIC FUNCTIONS

To obtain formulas for derivatives of trigonometric functions, it will be necessary to first prove several results about limits. Whenever we discuss limits of trigonometric expressions involving $\sin \theta$, $\cos t$, $\tan x$, and so on, we shall assume that each variable represents the radian measure of an angle or a real number.

Let θ denote an angle in standard position on a rectangular coordinate system, and consider the unit circle U in Figure 3.20. According to the definition of the sine and cosine functions, the coordinates of the indicated point P are $(\cos \theta, \sin \theta)$. It appears that if $\theta \rightarrow 0$, then $\sin \theta \rightarrow 0$ and $\cos \theta \rightarrow 1$. This suggests the following theorem.

FIGURE 3.20



Theorem (3.22)

$$(I) \lim_{\theta \rightarrow 0} \sin \theta = 0 \quad (II) \lim_{\theta \rightarrow 0} \cos \theta = 1$$

PROOF (i) Let us first show that $\lim_{\theta \rightarrow 0^+} \sin \theta = 0$. If $0 < \theta < \pi/2$, then, referring to Figure 3.20, we see that

$$0 < MP < \widehat{AP},$$

where MP denotes the length of the line segment joining M to P and $\widehat{AP}$ denotes the length of the circular arc between A and P . By the definition